Cody / CCP Communication Atlas

System map / commissioning, metadata, questions and one-way audio



CODY DEVICE

Native 2D UI, canonical snapshot, AFE microphone acquisition, retained pairing and encrypted audio transport.

WINDOWS CCP BROKER

Pairing journals, authenticated LAN presence, hook state, question tickets and optional audio receiver.

EXACT CODEX CHAT

Actual prompt/turn identity, native input and desktop approvals. No guessed chat or independent substitute session.

A saved relationship, a fresh reachable peer, a genuine followed chat and an accepted audio input are four different facts. Source code is not a deployment or physical-behavior certificate.

NOTES

Owners and transport boundaries

Each data source has one acquisition/authorization owner

OWNER	OWNED DATA / ACTION	BOUNDARY
Device snapshot / AFE	Real enhanced PCM, VAD and sensor observations; acquisition errors and timestamps.	Consumers never reread hardware or invent freshness. Device voice ring belongs to transport, not an alternate microphone.
Device CCP + UI	NVS active/pending association, canonical voice states, routes and visible controls.	Wake word and double tap request capture. No duplicate capture; current input prerequisites and safe-route admission apply.
PC CCP runtime	DPAPI journals, discovery/presence, hook ActivityStore, tickets, AES-GCM receiver and completed WAV.	Explicit questions have private capabilities. Unauthenticated data cannot establish presence or select the target chat.
Codex native endpoint	Exact thread/turn read, steering or start with localAudio; desktop-owned approvals.	Explicit loopback endpoint required. Installation, authorization, dependencies and current desktop-chat attachment remain unverified.

NOTES

One connection, three different facts

01 / Orientation

Pairing

Cody remembers one PC identity and a retained link key. A saved record does not prove that the PC is reachable.

Presence

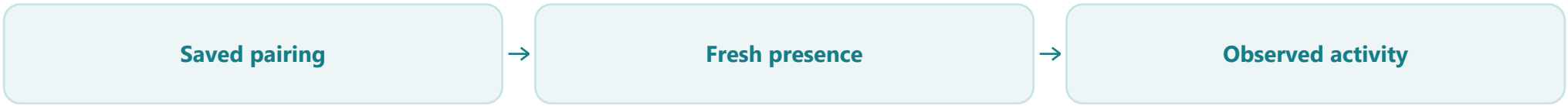
Authenticated current replies establish that the retained PC service is reachable on the local LAN. Public Internet is a separate condition.

Activity

Genuine Codex hook observations describe a chat and its work. An explicit question has its own ticket; its context label cannot invent a chat.

Read this atlas

Pages 1-2 map device, broker and native-chat ownership. Pages 4-7 cover commissioning and recovery; pages 8-13 preserve activity, questions, observations and trust. Pages 14-26 specify wire, API and audio contracts. The source crosswalk on page 27 and the final dated register retain earlier decisions and evidence.



Start from the saved state

02 / Boot and network / screens 01-11, 39-42

No Wi-Fi profile

A genuinely absent saved profile opens Welcome 04, installation 05-06, Bluetooth provisioning 07 and local connection 09. Debug can show handoff 39. Continue opens preferences 10, Personalization 36 and skin 11.

Saved Wi-Fi, no local IP

42 / No connection preserves credentials while the Wi-Fi owner reconnects. Network setup explicitly reopens 04-09; Later returns to Idle. Invalid or stale observations never count as a valid connection.

LAN ready

Without a saved Codex pairing, 40 offers Yes or No. No starts no CCP request and changes no settings. With a retained pairing, search the saved PC on 14; enter 15 only after current authentication.

Boot and provisioning boundaries

Cody AI Companion and A distraction-free sidekick sit above the full 192 x 219 floppy artwork at (137,164); title y=60, subtitle y=110. The 42-frame animation has a 3.5-second cycle; startup can extend wall time. Installation QR 06 starts no radio; Pair Cody starts Bluetooth 07. Its session QR exists only when ready to scan.

NOTES

Pair from the intended Codex PC

03 / First pairing / screens 12-15

Prepare the code

Hold the visible Cody still for 1.5 seconds, release, then choose Connect Codex. Preparing secure code / Please wait / About 30 seconds precedes the actual code. That text is an estimate, not a progress measurement.

Discover, then use the visible code

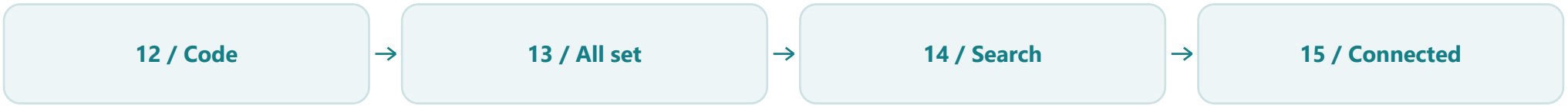
Ask Codex to discover Cody and pair using all six characters currently shown on the intended physical device. Use no old code from a screenshot or document. Zero or multiple matching offers is an explicit error; the first responder is never selected automatically.

Keep the code screen open

A request admitted by the service is not completed pairing. Wait for both sides to persist and confirm the relationship. The backend code window is five minutes; code-screen inactivity is three minutes from readiness.

All set -> connected -> actual work

13 / You're all set follows the persisted handshake. Let's go searches the saved PC immediately on 14 or reuses authenticated presence on 15. The old ten-second Idle detour is retired. Send a real prompt to establish activity; pairing does not manufacture hook events.



Commit the relationship on both sides

04 / Transaction ownership / CCP1

Prepare without destroying the active pair

The PC protects and persists its candidate with Windows DPAPI before PAIR. Cody writes a separate ccp_pending record. Each side preserves its previous active pairing until the authenticated commit is established.

Prove the candidate and the commit

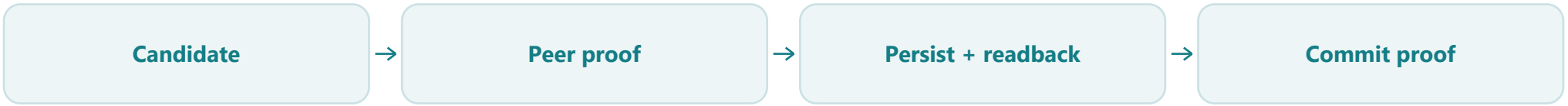
CONFIRM or signed READY proves the PC and permits Cody's commit/readback. READY now carries a fresh random challenge and the observed device address/port. Cody answers COMMITTED for that endpoint. Only the fresh matching response admits broker presence, promotion and current data; FIND alone does not. Lost acknowledgements and restarts can recover through another challenge.

Cancel and replace are different

Cancel discards Cody's candidate and preserves the operational old link. The PC candidate can remain pending. Replacing it requires a new visible code and explicit replace_pending:true. Possession of the old key does not authorize deleting the candidate.

Failures remain real

Storage corruption, failed writes/readback and ambiguous state are explicit failures, not missing settings. No silent repair, default substitution or success is reported. Pairing paused and Pairing error retain the incomplete candidate or storage cause.



Recover the same PC

05 / Setup, office networks and standby / screens 14-16, 32, 35, 41-42

One identity on a busy network

Even among 100 PCs, Cody searches the retained relationship. Other responding PCs remain outside it. Keep link resumes discovery; Pair again explicitly permits replacement; Forget requires a second confirmation.

Four independent outages

Public Internet down with a working LAN does not break local CCP presence. Missing Wi-Fi/IP preserves credentials and the pair. Missing PC/service preserves identity and retries discovery. A terminated or unavailable chat is distinct from loss of the paired PC.

Standby and wake

Ordinary connected standby preserves the route and pairing and is designed to retain notification reception. An authenticated notice requests wake through the power owner. Deep sleep, wake-word listening and radio retention are separate service decisions; physical recovery must be observed independently.

A deliberate escape

A stationary character hold opens Setup only from eligible resting routes; movement over 12 px, gaps over 250 ms, rotation or interruption cancels it. A hold spanning wake starts only after a fresh sample and restored artwork. Busy operations and pending cleanup cannot be bypassed. A normal application flash preserves NVS; erase and damaged storage are separate cases.

NOTES

Describe work without inventing it

06 / Hook activity / screens 16-20 and 32

Real observations

Reviewed/trusted plugin hooks observe the actual prompt and tool lifecycle. The current route follows the latest genuine prompt automatically; the old manual chat-selection proposal is historical. The shared local PC runtime can serve several open chats while retaining pairing.

Meaningful text

Working, Editing code and Running command describe observed phases. A phrase such as Building firmware needs a concrete operation or explicit agent progress. Project and branch labels are context; Reply ready means a response is available and does not certify task success.

Independent freshness

Presence normally polls every two seconds and expires after eight seconds without current replies. Hook activity expires after 120 seconds. An old observation is unavailable/stale, not new work. Paused, approval waiting and session ended retain their distinct meanings.

Native approvals remain on the computer

PermissionRequest can produce On your PC; Cody does not grant native tool permissions. Explicit semantic questions and MCP notifications have their own identities and cannot create UserPromptSubmit, a fabricated session or an authorization.

NOTES

A choice is delivered only after its recipient ACK

07 / Questions / screens 26, 27 and 31

Read, choose, review

26 / Codex needs you shows the stated context, title and body. Open reveals two cards, or the pair plus one centered third card. Stable choice IDs preserve meaning. A selection opens Review choice; only Send transmits. Back returns without answering.

Pending -> Sending -> received -> delivered

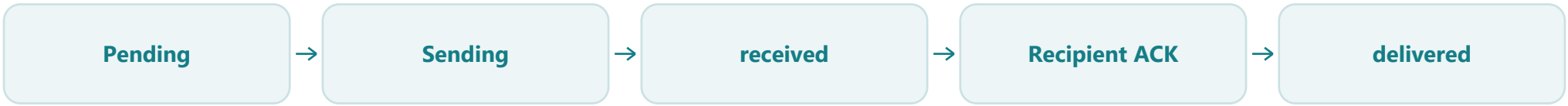
The local device keeps the chosen answer fixed while sending. received means the broker recorded it. The caller polls its private ticket/capability and obtains answer_id; a subsequent matching acknowledge_answer_id proves recipient receipt. Only delivered produces Reply sent.

Expiry never chooses an answer

A question lasts at most 180 seconds; duplicates, reopening a view and reconnecting do not renew it. Before Send show Request expired; after Send show Reply unconfirmed / Check on your PC when delivery is uncertain. Cancelled/rejected and expired requests stay terminal.

Consecutive requests and offline recovery

The broker waits for the device to acknowledge closure before replacing a question. An active question stays ahead of routine activity and battery updates. Repeating the same answer is idempotent; another answer cannot overwrite it. Offline preserves the question and its original deadline.



PR/check observations stay with the work

08 / Decision A03 / screens 21-24; screen 25 retired

Watch the explicitly supplied PR head

21 / PR pending displays the indicated PR and exact head from the caller's snapshot. A Stop hook does not prove check status. Cody does not merge, approve, close, rerun or comment on a PR.

Only complete current check data supports totals

22 / Checks running or Checks passed uses numeric totals only from a complete check list for the same head. Checks passed has no buttons; after the narrative and three seconds the prior work view resumes.

Show names, not logs

23 / View failures opens real failed-check names: at most six, each at most 48 printable ASCII characters. Unrepresentable input is rejected instead of truncated. 24 shows three names per page with up to two lines each, vertical scroll and lateral Back.

Keep diagnosis on the PC

Check rows are not selectable actions. View log and diagnostic screen 25 are retired. Logs, traces and repairs remain on the computer. Screens 26/27 are not a substitute PR action channel. Screens 28-30 originated as a future voice route. Source version 0.6.5 now adds one-way capture, transport and native input; installation and physical delivery remain unverified, as qualified in the audio chapter.

NOTES

Settings, gestures and consistent controls

09 / Local preferences / screens 10, 11, 36 and 41

Options returns to its caller

Setup shows Connect Codex and Network setup as a pair, with Options centered below. Options opens Make it yours, Personalization and Cody's look, then returns to the correct caller. Local preferences do not replace the paired PC or erase the network.

Persist before confirming

Shake to wake, Auto-rotate and Sounds are On/Off settings. Orientation is Auto while auto-rotate is on; manual 0 or 180 is available while it is off. Sounds Off stops playback without disabling microphone/wake word. Skins use stable IDs: Pearl Mist, Aurora Veil, Cosmic Stardust, Enchanted Mosaic, Dreamwave. Write/readback failure preserves the requested value and original error.

Action language

Two actions use semantic gradient cards side by side; three add one centered card below; one uses an icon-left wide capsule. Labels have no teletype. Controls appear and admit touch together after title/body. Back remains left at (10,203), 60 x 60. Protect each circular chord and every QR quiet zone.

Physical buttons remain separate

PWR standby/wake follows build policy; Debug can expose screenshot/restart gestures. BOOT restores a sleeping panel or toggles its existing display mode; its Debug long hold restarts. Neither normal restart nor entering Setup removes saved Wi-Fi/pairing. Explicit erase is a separate user operation.

NOTES

Trust boundaries and security backlog

10 / Security / source review, not a production certification

Authentication is not confidentiality

CCP1 uses HMAC with identity/nonce checks. Payloads remain readable on the LAN; CRC detects storage corruption without encrypting a key. Windows DPAPI protects PC persistence. The six-character code is not sent in clear, but the captured verifier transcript permits offline guessing; PBKDF2 alone is not a PAKE.

Private capabilities and explicit destructive intent

MCP ticket/capability values stay on the PC. Signed reset/erase admission does not prove completion. Restart preserves configuration; factory reset removes preferences, Wi-Fi and pairing while retaining firmware, assets and SD, and reboots only after verified erasure. The nine-digit anti-mistake confirmation is not a password and must not be filled from documentation.

Source hardening in this revision

Endpoint-bound one-time READY/COMMITTED challenges prevent recorded FIND from selecting a current-data peer. Firmware receive-pass and global/per-source ingress budgets protect maintenance. PC discovery and PBKDF2 work have explicit limits; rejected-traffic summaries and rotating logs bound diagnostics. The firmware and plugin must be updated together.

Still requires design and evidence

A standard PAKE or high-entropy out-of-band pairing plus authenticated encryption needs a separate protocol decision. Do not call a LAN trusted merely because it is local. Firmware compilation, adversarial checks and physical pairing/recovery evidence are separate from this PDF consolidation.

NOTES

Keep evidence and open decisions visible

11 / Working notes / carried forward from the connection atlas

Evidence levels

Source implementation, static cross-component review, host assertions, successful build/flash and physical observation are different claims. This atlas is a source-based document; screenshots/schematics are not current device evidence. done.md records dated results.

Earlier observed failure retained

A prior authorized capture showed Yes at 16497 ms, Connect Codex at 16498 ms, then a CST9217 report of fifteen FF bytes with ACK=FF at 17112 ms (expected AB). ESP_ERR_INVALID_RESPONSE stopped rendering. The optional runtime-acquisition correction does not establish the hardware root cause.

Scope of the touch correction

After startup only, an invalid touch report can interrupt gestures while rendering continues. The acquisition owner keeps the actual error, clears queued gesture reports and requires a genuine acquired release. No release is fabricated; other startup/locking/route failures remain terminal.

Use the review worksheet

Record the observed device route, PC/service state, connection flow, decision to retain and open question. Preserve screen IDs across repagination. A failure or partial failure is not a successful delivery, current sensor reading or permission.

NOTES

Ports and packet envelopes

Transport is separate from UI, chat identity and Internet reachability

CHANNEL	SOURCE CONTRACT	BOUNDARY
UDP 42117 / device	Pairing discovery and normal CCP1 endpoint. The retained device identity is a MAC address encoded as 12 lowercase hexadecimal characters.	The device acquires Wi-Fi and an IP address first. A PC identity contains 32 random lowercase hexadecimal characters.
UDP 42118 / PC broker	Normal CCP metadata and commands use printable ASCII, canonical | field separators and HMAC proofs.	Ordinary packets: 512 bytes. ACTIVITY/TECH: 1024 bytes. EVENT: 1280 bytes. No application fragmentation.
UDP 42119 / optional PC audio	Binary CYA1 device packets and CYA2 authenticated ACKs use AES-256-GCM.	49-byte authenticated header + up to 960 PCM bytes + 16-byte tag = 1025 bytes maximum. Audio has a separate ingress owner.
Loopback PC control / native input	Authenticated local control: 32 workers, a 3-second socket deadline and a 64 KiB body. Native audio uses an explicitly configured loopback WebSocket endpoint.	URL credentials, query strings and fragments are rejected; a literal loopback IP is required. Configuration does not prove attachment to the desktop chat.

NOTES

Commissioning packets and persistence

512-byte CCP1 envelopes; the visible six-character code is never transmitted

MESSAGE / DIRECTION	CANONICAL FIELD ORDER	MEANING
DISCOVER / OFFER	request request -> device -> session -> salt -> tag	The PC discovers; the device offers. The six-character code admits exactly one verifier match. PBKDF2-HMAC-SHA256 uses 50,000 iterations, a 16-byte salt and a 32-byte verifier.
PAIR	device -> session -> pc -> port -> nonce -> proof	The PC persists its DPAPI candidate first. Link key = HMAC(verifier, link|device|session|pc|nonce). The device writes a separate NVS ccp_pending record.
PAIRED	device -> session -> pc -> nonce -> proof	Device -> PC: pending write succeeded, not active new pairing. Both old active records remain until commit.
CONFIRM / ACTIVE	device -> session -> pc -> nonce -> proof device -> session -> pc -> nonce -> proof	PC peer proof permits device commit/readback. ACTIVE confirms that stage. Broker promotion still requires the fresh endpoint challenge.

NOTES

Rediscovery proves the observed endpoint

FIND -> READY -> COMMITTED precedes current ACTIVITY / TECH / EVENT

MESSAGE	CANONICAL FIELD ORDER	ADMISSION RULE
FIND	device -> pc -> nonce -> activity -> events -> current -> state -> proof	Current source requests activity=4 / events=2; current and state describe the open device question. HMAC domain binds exact capability fields.
READY	device -> pc -> nonce -> challenge -> address -> port -> proof	The broker supplies a fresh random challenge, the observed device IPv4 address and its UDP port. The ready-v2 proof binds that endpoint; lifetime: 2 seconds.
COMMITTED	device -> pc -> nonce -> challenge -> address -> port -> proof	committed-v2 proof from that exact endpoint consumes the matching challenge once. Only then admit presence, commit recovery and current data.
Retained link versus fresh proof	DPAPI active/pending PC records; NVS ccp_pairing/ccp_pending	FIND alone cannot establish presence, redirect notifications or release data. Retained pairing alone cannot admit audio START.

NOTES

ACTIVITY and coherent chat identity

1024 bytes; one current snapshot, no transcripts and no manufactured hook

PROJECTION	FIELDS / LIMITS	OWNER / VALIDITY
ACTIVITY order	device -> pc -> nonce -> chat -> revision -> phase -> project -> description -> branch -> proof	Authenticated after fresh COMMITTED. chat=SHA256 actual followed session ID; revision is volatile diagnostics, not durable ordering.
Human metadata	project/branch: at most 64 printable ASCII characters; description: at most 192. Text is hex encoded; - denotes unavailable.	Only working/editing/running admits a description. The branch owner reads the nearest checkout's .git/HEAD or worktree pointer; DETACHED is explicit. Rejected text is not truncated.
No chat / no new work	zero chat; phase=none; project=-; description=-; branch=-	The latest genuine UserPromptSubmit selects context. Tool/Stop events from background chats cannot steal it. Hook activity expires after 120 seconds; presence expires after 8 seconds.
Paired ACTIVITY + TECH	One ActivityStore lock, same chat and revision; independent consumed-nonce flags	TECH is admitted only after matching ACTIVITY. No merge with an older projection or freshness fallback; next FIND acquires another coherent pair.

NOTES

TECH: report sources, never infer token usage

1024 bytes; technical-v2 authenticated order is exact

FIELD GROUP	SOURCE / FORMAT	UNAVAILABLE AND EXPIRY
Exact TECH order	device -> pc -> nonce -> chat -> turn -> revision -> model -> effort -> modelage -> compaction -> compactseq -> compactage -> activityage -> turnelapsed -> used -> capacity -> cumulative -> input -> output -> threshold -> usageage -> cost -> costage -> costttl -> proof	Lowercase hex identities/model; decimal ms without leading zeroes; - unavailable. All fields and their order are authenticated.
model / modelage / effort	Full model slug: 1-128 ASCII characters, codes 33-126. Model age follows genuine hook emission. effort=-.	A reported model does not prove rerouting. The model expires after 120 seconds; an absent model requires an absent age. Reasoning effort has no connected source.
compaction / compactseq / compactage	Real PreCompact/PostCompact on exact chat/turn; none,active,refreshed,orphaned,interrupted,stale.	Only active age denotes observed start; other ages are transition ages. Duplicate/old-turn boundaries rejected. Refreshed never creates token data.
activityage / turnelapsed / usage	Genuine emission age and observed duration since the prompt. Duration freezes at Stop, Interrupt or SessionEnd.	used, capacity, cumulative, input, output, threshold and usageage must all remain -. No guessed occupancy, model-global denominator, progress percentage or free-token bar.

NOTES

Catalog cost is a qualitative local-day lease

TECH cost/costage/costttl; numbers and billing stay on the PC

FACT	CONTRACT	EVIDENCE BOUNDARY
Band	unknown / economy / balanced / expensive, matched to the exact observed model.	Qualitative public Standard catalog classification; no currency, bill or inference-cost claim.
Original age	costage is original public catalog retrieval age. Repeated packets never refresh that source age.	Separate from hook emission, receiver transport timestamps and PC heartbeat.
Lease	costttl lasts until the next PC local midnight. Duration is positive; catalog age and expiry are at most 26 hours to allow daylight-saving transitions.	Firmware expires the lease using its own monotonic elapsed time, without comparing wall clocks. Unknown requires both durations to be -.
Independent invalidation	Require model age below 120000 ms. A stale model or transport, an expired lease or a disconnect invalidates cost.	Catalog source, rates, retrieval attempts, errors and classification thresholds remain PC-owned. An unavailable catalog is never fabricated.

NOTES

EVENT, replies and recipient closure

Immutable ticket; 180-second maximum; received is not delivered

MESSAGE / PAYLOAD	ORDER / BOUNDS	RULE
EVENT	device -> pc -> nonce -> id -> state -> ttl -> data -> proof	1280-byte envelope; ttl is 0-180 remaining seconds, never UTC. Pending/received require at least 1 second. The current nonce admits one publication; repetition never renews the deadline.
REPLY / REPLYACK	device -> pc -> nonce -> id -> choice -> proof device -> pc -> nonce -> id -> choice -> status -> proof	512-byte packets; stable choiceID; same answer idempotent, another cannot overwrite. Broker receipt is not recipient confirmation.
EVENTACK	device -> pc -> nonce -> id -> state -> proof	Terminal closure acknowledged before publishing successor. Unknown ticket after broker restart produces unavailable/rejected; never reconstruct an answer.
data v1 order	version, kind, context, title, body, choice_count, choices; each choice retains id, label and detail	ASCII lengths: context 0-24, title 1-24, body 1-63. Questions have 2 or 3 choices; other events have none. Choice label: 1-12; detail: 1-63. Excess or unrepresentable values remain errors.
Private PC capability	Random 128-bit ticket and 256-bit capability. Poll and cancel require both; acknowledge_answer_id confirms receipt.	The capability never leaves the PC. Explicit caller context never invents verified hook/chat identity or native authorization.

NOTES

Signed controls keep user intent explicit

Clock, preferred name, Debug and reset are independent from audio transport

OPERATION	FIELDS / OWNER	ADMISSION / OUTCOME
CLOCK / CLOCKACK	device -> pc -> nonce -> id -> op -> timezone -> enabled -> start -> end -> proof device -> pc -> nonce -> id -> op -> status -> error -> initialized -> timezone -> enabled -> start -> end -> proof	read sends all settings as -1; init adopts a supported host timezone once; set sends explicit values. UTC/Rome/Ho Chi Minh IDs: 0/1/2. Minutes: 0-1439; quiet hours use a half-open window.
NAME / NAMEACK	NAME: device -> pc -> nonce -> id -> op -> name -> proof. NAMEACK adds status -> error -> available before name. NAMECAP: device -> pc -> nonce -> name=1 -> proof.	Read/set permits at most 24 printable ASCII characters without padded edges. Empty explicitly clears. Persist and read back before applied status. Fresh endpoint-bound capability is single-use.
DEBUG / DEBUGACK	DEBUG: device -> pc -> nonce -> id -> mode -> scroll -> proof. DEBUGACK adds status -> error -> count before proof. DEBUGCAP: device -> pc -> nonce -> debug=1 -> proof.	Only Debug firmware advertises this single-use capability. Selection does not choose a chat or read hardware independently. pixels_verified remains false; a new command needs a fresh capability.
RESET / FACTORYRESET	device -> pc -> nonce -> id -> code -> proof device -> pc -> nonce -> id -> code -> proof	Current user intent and an exact nine-digit confirmation are required. Restart retains configuration. Factory reset removes Wi-Fi, pairing and preferences; firmware, assets and SD remain.
RESETACK / FACTORYRESETACK	device -> pc -> nonce -> id -> status -> proof device -> pc -> nonce -> id -> status -> proof	accepted proves command admission, not boot or erase completion. Fresh presence proves reboot; verified erase precedes the factory-reset reboot.

NOTES

MCP calls and their local authority

Method names are API boundaries; identity and permissions stay typed and explicit

CALLS	PURPOSE	AUTHORIZATION / LIMIT
ccp_discover_devices / ccp_pair_device	Bounded device discovery; visible-code pairing with explicit pending replacement.	Zero or multiple matching offers fail. A simulated identity must be explicitly included and visibly labelled SIMULATED.
ccp_connection_status / ccp_list_sessions / ccp_report_progress	Read retained/current state, list observed sessions and explicitly report a work descriptor.	Never manufacture hooks or choose a chat from a label. The latest actual prompt selects the followed context.
ccp_publish_notification / ccp_ask_question / ccp_poll_request / ccp_cancel_request	Explicit notice/question; capability-bound poll/cancel; recipient answer ACK.	Native approvals stay exclusively on the PC. This reference PDF does not itself authorize messages.
ccp_read_name / ccp_set_name / ccp_read_clock / ccp_configure_clock / ccp_debug_screen	Bounded preference reads/writes and a requested Debug display.	Readback precedes applied status. Unsupported timezones and persistence failures retain their real cause.
ccp_reset_device / ccp_factory_reset_device / ccp_audio_status	Explicit restart/factory-reset intent; audio receiver diagnostics without PCM or secrets.	Audio status neither chooses a chat nor starts recording. Reset confirmation comes only from the current user message.

NOTES

Initial resource budgets and failure ownership

Source constants bound work; they are not throughput or anti-jamming guarantees

OWNER	INITIAL LIMIT	FAILURE / EVIDENCE
Firmware ordinary UDP	8 packets or 5 ms per receive slice; global rate 32/s, burst 64; per IPv4 rate 16/s, burst 32; 16 source slots.	Maintenance still runs. A slice cannot preempt blocking storage or crypto. Runtime tuning and stack headroom remain unverified.
PC ingress / discovery	Global rate 64/s, burst 128; per IP rate 16/s, burst 32; 64 slots. Discovery: 16 offers, 4 per IP, 128 packets.	Overflow discards the entire result. Pairing verification allows 16 PBKDF2 candidates within 2 seconds; no partial association staging.
Challenges / control / logs	128 challenges, 2-second lifetime. Local control: 32 workers, 3-second deadline, 64 KiB body. Rejection summary every 5 seconds; broker log: 3 x 2 MiB.	Original source diagnostics and errors survive. Startup runtime.log is a separate logging boundary; no replay or denial-of-service runtime certification.
Audio / PC artifacts	4 active captures/jobs; 64 transient utterance records; 60-second, 1,920,000-byte PCM ceiling per capture; 10-second abandoned-capture timeout.	Disk quota: 64 MiB and 1024 WAV files. Reject new persistence; never silently delete another utterance. Whole native submission deadline: 15 seconds.

NOTES

Audio is a separate one-way binary channel

Plugin 0.6.5 source increment / UDP 42119 / no device playback or full duplex

BINARY HEADER OFFSET	FIELD / WIDTH	MEANING
0	magic / 4 bytes	CYA1 comes from the device; CYA2 comes from the PC. Network-order struct !4s6s16s16sIBH defines the 49-byte header.
4 / 10 / 26	device / 6 bytes; utterance / 16 bytes; discovery nonce / 16 bytes	The 12-character hexadecimal device identity becomes binary. A fresh authenticated peer and discovery nonce admit START; an active utterance remains bound to its original endpoint, credential and nonce.
42 / 46 / 47	sequence / uint32; kind / uint8; plaintext length / uint16	START=1: sequence 0, no PCM. DATA=2: at most 960 bytes. END=3: no PCM. CANCEL=4: a new sequence/nonce, no PCM. Audio is little-endian PCM16, mono, 16 kHz.
49 + plaintext length	AES-256-GCM ciphertext + 16-byte tag	The whole header is AAD. Maximum: 49 + 960 + 16 = 1025 bytes. Key = HMAC-SHA256(link_key, cody-audio-v1|utterancehex|discoveryhex).
AEAD nonce / ACK	DAT1||uint64(sequence); ACK1||uint64(ackcounter)	A retry repeats identical ciphertext; never reuse a nonce with different plaintext. PC ACK counters strictly increase. ACK plaintext !IB contains the highest contiguous received device sequence and status.

NOTES

A ten-second ring tolerates a stalled PC

160000 samples x 2 bytes = 320000 bytes PSRAM; metadata and transport buffers are additional

STAGE	DEVICE / PC OWNER	BOUNDARY
Capture and display	Existing AFE enhanced mono 16 kHz PCM and VAD. Wake word or double tap requests Listening; capture starts with the next real AFE block, without preroll. At the same lower position, alternate exact ASCII captions "Shake to fade" and "We will send the audio to Codex for transcription"; scroll the long caption inside the clipped row.	Hide Cody. Synthetic wave: four phase-shifted colored curves in a 440 x 40 band; 16 poses at 8 FPS, Q8 antialiasing and alpha overlaps, no runtime trig or voice-amplitude tracking. One row cycles model -> cost band -> thinking effort -> branch -> project, skipping unavailable values with existing fades/scrolling. No fixed cost pill. Elapsed and remaining seconds show the 60-second limit.
Ring and sender	Keep PCM until cumulative authenticated ACK. Four DATA packets, each 480 samples / 960 bytes. Retry identical packets every 100 ms; transport every 5 ms, at least one tick.	Ten-second PSRAM/grace survives CCP loss only for the same authenticated key and IP. Overflow fails; no overwrite or SD/flash fallback. Capture ends at 1.2-second speech silence or 60 seconds.
PC receiver / persistence	Freeze the genuine START chat. Discard out-of-order DATA; ACK the contiguous prefix. Persist completed 16 kHz, 16-bit mono WAV under private runtime artifacts.	Separate UDP/native workers: 4 captures/jobs, 64 records, 60-second PCM limit. Disk quota: 64 MiB/1024 WAV. START requires recent readiness for the followed chat.
Pause / deliberate shake	Shake cancels Listening/Sending; no voice Cancel button. Microphone pause stops capture/transport. Transport failure after transmitted END is Unconfirmed.	No automatic wake/restart. CANCEL cannot revoke native submission in flight; retain its actual outcome. The hint is not proof of transcription.

NOTES

PC receipt, native acceptance and work completion

Native endpoint is an explicit integration dependency, not a discovered guarantee

ACK / NATIVE STEP	SOURCE BEHAVIOR	HONEST OUTCOME
Audio ACK statuses	1: START accepted. 2: PCM received. 3: whole utterance received. 4: native Codex accepted. 5: failed/cancelled. 6: native input unconfirmed.	END receipt is not native acceptance. Native acceptance is not task completion. Unconfirmed submissions never repeat automatically.
Read-only native readiness	CODY_CODEX_AUDIO_ENDPOINT selects an existing literal-loopback WebSocket. Probe initialize and exact thread/read within 5 seconds; refresh every 5 seconds, freshness limit 10 seconds. Reject authenticated START without recent readiness for the followed chat.	The probe does not prove localAudio acceptance, model support or transcription. cryptography and websocket-client remain required; optional token file: CODY_CODEX_AUDIO_TOKEN_FILE. Never expose token values.
Native localAudio	input=[{"type":"localAudio","path":completedWAV}]. Preserve the START chat; resolve its current hook turn at submission. Active turn: turn/steer(expectedTurnId); otherwise thread/resume then turn/start. clientUserMessageId is the recording hex ID.	Revalidate followed-chat identity and cancellation. Concurrent native turn conflicts fail explicitly; retain the WAV and never retry automatically. The correlation ID provides no idempotency guarantee.
Native approvals and uncertainty	The whole submission has a 15-second monotonic deadline. Authorization and user questions stay on the desktop. Status exposes recording ID, native turn, WAV artifact and uncertainty.	Explicit rejection means Failed. Uncertain submission means status 6: Check on your PC. Endpoint attachment, installation, account/backend, dependencies and physical audio remain unverified.

NOTES

Source crosswalk

Original pagination and screen identities retained as references

Current concepts and trust	Service 1-2, 7-10; Connection 1, 9-11, 13	1-3, 8-12
First pairing and persistence	Service 3-6; Connection 2-8, 12	4-7, 15-16
Setup decision and recovery	Service 11-15; Connection 15-19	4, 7, 11 + grid
UI decision and touch cause	Service 16-18; Connection 20-22	11, 13 + grid
Earlier activity proposal	Service 19-20; Connection 23-24	8, 17-19 + grid
Retained pairing revision	Service 21-24; Connection 25-28	4-7, 16 + grid
Notifications and PR revision	Service 25-28; Connection 29-32	9-10, 20 + grid
Boot, settings and controls	Service 29-32; Connection 33-36	4-5, 11, 21-22
Sketch and open decisions	Connection 14	13, worksheet

Service means the former 32-page CCP Service Atlas. Connection means the former 36-page Codex Connection Atlas. All substantive topics and dated evidence are carried forward once, in English. Repeated preservation boilerplate is consolidated here. Earlier diagrams remain in Git; no historical illustration becomes proof of current physical behavior.

Communication review worksheet

Sketch the next refinement / original Connection page 14

DEVICE / observed route and input

PC / CODEX / service and chat

CONNECTION FLOW / identities, messages and timing

DECISIONS TO KEEP

OPEN QUESTIONS

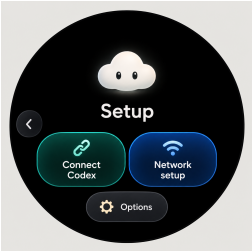
Revision register

Original dates, screen IDs, decisions and evidence are retained. Source code does not prove device behavior.

DATE / TYPE	DESCRIPTION / ORIGINAL PAGE AND SCREEN IDS
2026-10-04 CHANGE	**One intended PC on a busy LAN** Affected: Original Connection pages 9 and 14; screens 12-16, 32, 35B, 40-42 Cody retains one explicit PC identity even among 100 PCs; other responders do not replace it. First commissioning, a persisted relationship and authenticated current presence are separate facts. The original connection worksheet remains a separate writing page. Evidence / limits: Original design reference, not a current capture.
2026-10-04 CHANGE	**A03: PRs remain read-only** Affected: Service scope/security; Connection 9-14; screens 21-30 Retain PR/check states and the indicated head, show only real failed-check names, and leave diagnosis/actions on the PC. Screen 25 is retired. Native approvals remain computer-owned; voice screens 28-30 are future work. Evidence / limits: Original design decision; not a physical capture.
2026-10-09 ADDENDUM	**Startup decisions and reversible Setup** Affected: Service 11-13; Connection 15-17; original screens 12-16, 40-42 The earlier analysis omitted saved Wi-Fi without Codex pairing and a reversible Setup entry. Add 40 / Yes-No, 41 / character hold and 42 / reconnect wait. No/Later causes no persistence or new CCP request. Saved pairing searches 14 and requires current authentication for 15. PWR/BOOT do not clear saved pairing. A factory erase remains separately authorized. Evidence / limits: Source/static review only in this revision; firmware, host and physical verification remained pending.
2026-10-09 ERRATA	**Setup recovery after standby** Affected: Service 14-15; Connection 18-19; screens 04-06, 12, 35B, 41 The legacy Welcome timeout path obscured reopening Setup. A resting Welcome tap release or eligible stationary hold can reopen 41. Wake time cannot count toward the hold until fresh acquisition and visible restoration; movement, gaps and rotation cancel it. Busy provisioning and pending cleanup are excluded. Evidence / limits: A 90-second capture showed LAN/DHCP, inactivity return and standby, but no Connect Codex press. Corrected physical behavior was still pending.

Revision register

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DATE / TYPE	DESCRIPTION / ORIGINAL PAGE AND SCREEN IDS
2026-10-09 CHANGE	**Approved action layout and Options entry** Affected: Service 16-18; Connection 20-22; screens 05-15, 35B, 36, 40-42 Two semantic gradient cards sit side by side; a third is centered below. A single action is an icon-left wide capsule. Back stays lateral, render/touch share bounds, and QR quiet zones are protected. Options reopens preferences and skin before returning to Setup. Evidence / limits: Approved design reference, not a device screenshot.  Approved multi-action reference / 09 October 2026.
2026-10-09 ERRATA	**Malformed touch report interrupted rendering** Affected: Service 17-18; Connection 21-22; screens 12 and 35B The original analysis failed to distinguish an optional runtime touch error from renderer failure. A capture recorded Yes at 16497 ms, Connect Codex at 16498 ms, then fifteen FF report bytes and ACK=FF at 17112 ms. The expected ACK was AB; ESP_ERR_INVALID_RESPONSE stopped rendering. The owner preserves the original error. Only after startup, invalid acquisition may interrupt gestures while rendering continues, with queued reports discarded and a real release required. The FF hardware root cause was not established. Evidence / limits: Measured failure retained; corrected physical touch behavior remained pending.
2026-10-10 ADDENDUM	**Earlier CCP 0.3 activity proposal** Affected: Service 19-20; Connection 23-24; original Service 1-2, 7-8, 10 / Connection 10-11, 13 Metadata-only hooks, shared local runtime, explicit manual session selection and nonce-bound ACTIVITY were the proposal recorded then. Screen 15 proved PC presence only; 18U made unavailable activity explicit. Two-second polling, eight-second presence and 120-second hook freshness were separate. Historical limitation: branch acquisition, notifications and voice were then future work. Manual chat selection was later retired in favor of the genuine latest prompt; later source revisions supersede that policy. Evidence / limits: User-reported Connected applied to earlier firmware/CCP 0.2, not proof of the new 0.3 activity proposal.

Revision register

Original dates, screen IDs, decisions and evidence are retained. Source code does not prove device behavior.

DATE / TYPE	DESCRIPTION / ORIGINAL PAGE AND SCREEN IDS
2026-10-10 CHANGE	**Retained pairing and two-sided commit** Affected: Service 21-24; Connection 25-28; screens 04-16, 32, 35B, 36, 40-42 Separate commissioning, recovery and real chat activity. Already paired offers Keep pairing, Pair again and explicitly confirmed Forget. Retry keeps the same PC; Cancel abandons the candidate; Forget removes only Cody's CCP records. Preferences and Wi-Fi remain separate. PC DPAPI candidate and Cody ccp_pending preserve old active records. CONFIRM/fresh READY proves the peer; persisted ACTIVE/COMMITTED proves promotion. Repeat COMMITTED on each current READY to recover lost acknowledgement. Explicit replacement is required for an existing pending PC candidate. Evidence / limits: Source revision; paired Internet outages, restarts and wake behavior require their own device evidence.
2026-10-10 ADDENDUM	**Notifications, choices and recipient delivery** Affected: Service 25-28; Connection 29-32; screens 21-27, 31-32 26 opens a bounded explicit ticket; 27 uses cards plus a separate Send review. The recipient must acknowledge answer_id before delivered / Reply sent. The 180-second deadline applies even to received; neither disconnection, timeout nor cancellation chooses a reply. Terminal states do not reopen. The recorded wire revision used activity=1/events=1 and a 1280-byte EVENT ceiling, versus 512-byte ordinary packets. Context/title/body and choice text are bounded printable ASCII; capabilities stay private on the PC. Later capability revisions must be read from current source, not this historical example. Evidence / limits: Source contract; original 24/24/63-character context/title/body and 12/63 label/detail limits are historical revision facts.
2026-10-10 ERRATA	**Acknowledgement is admission, not reset completion** Affected: Service 27-28; Connection 31-32; reset and factory-reset commands Restart retains configuration; factory reset removes preferences, Wi-Fi and pairing but preserves firmware, assets and SD. Destructive intent and its nine-digit anti-mistake confirmation must come from the current user message. Never complete it from a document. Signed ACK means accepted; new presence proves reboot, and verified erasure precedes factory-reset reboot. Evidence / limits: Real erase/boot outcomes remain distinct from authenticated command admission.

Revision register

Original dates, screen IDs, decisions and evidence are retained. Source code does not prove device behavior.

DATE / TYPE	DESCRIPTION / ORIGINAL PAGE AND SCREEN IDS
2026-10-10 CHANGE	**Current boot, setup and complete controls** Affected: Service 29-32; Connection 33-36; boot 01-03, Wi-Fi 04-09/42, preferences 10/36/11, Codex 12-15/40/41 Boot gains the title/subtitle above complete floppy art. Installation QR starts no radio, Bluetooth session QR exists only when ready, and local Wi-Fi/IP gates Continue. Code preparation estimates do not fabricate progress. All set searches immediately, removing the old Idle detour. Four local preference rows, stable skin IDs and readback-confirmed saves retain real errors. Controls appear together after narrative completion, without teletype labels. Historic artwork sizes and obsolete route sketches remain dated references, not the current contract. Evidence / limits: Static source geometry and approved art only; this document revision executed no simulator or physical-device validation.
2026-10-10 CHANGE	**Consolidated service and connection atlas** Affected: Both source atlases in full; consolidated pages 1-13 and Revision register Merge the former 32-page service atlas and 36-page connection atlas into one English reference. Deduplicate shared setup/notification history, retain original page/screen crosswalks and the review worksheet, and replace scattered closing errata with this reusable dated register. The register uses a fixed Date/Type column and a separate narrative/artwork column, repeating its header across pages. Supported categories are Security, Errata, Addendum and Change. Old source documents remain recoverable in Git; their historical evidence and qualifications are preserved above. Evidence / limits: PDF authoring and visual review are document checks, not firmware/security runtime validation.
2026-10-11 ERRATA	**English correction retained from both earlier atlases** Affected: Service page 33; Codex Connection page 37; original historical pages referenced therein Service original page 33 and legacy Connection original page 37 translated historical Italian headings, diagram captions, body copy and footers into English. Original dates, screen IDs, technical values, artwork meaning and evidence boundaries were preserved. This dated correction is now consolidated in the canonical final grid. Evidence / limits: Document language and visual review only; no firmware or physical validation.

Revision register

Original dates, screen IDs, decisions and evidence are retained. Source code does not prove device behavior.

DATE / TYPE	DESCRIPTION / ORIGINAL PAGE AND SCREEN IDS
2026-10-12 CHANGE	**One canonical CCP communication reference** Affected: All three technical source PDFs; A5 quickstart context; current CCP1.md, ccp_voice_input.py and LUNASHELL_voice_input.cpp Unify the existing CCP Connection Atlas, former Service Atlas and legacy Codex Connection Atlas into coherent system, commissioning, wire/API, audio and evidence chapters. Keep the separate four-page A5 pairing quickstart as the nontechnical user guide. Preserve earlier meaningful decisions, original page/screen crosswalks and dated Errata, Addendum, Change and Security records. Current packet fields, order and ceilings; hook/technical projections; clock/name/debug/reset; MCP calls; and resource budgets are source-derived. Metadata remains authenticated plaintext. Audio is a separate AES-256-GCM source increment and does not strengthen six-character commissioning. Voice documentation follows source version 0.6.5 and the 12 October UI decision: enhanced PCM from the next real AFE block after wake detection, a ten-second PSRAM ring, authenticated UDP 42119, frozen native thread/turn identity and localAudio input. Capture lasts at most 60 seconds; deliberate shake cancels Listening or Sending without a voice Cancel button. Hide Cody. The upper synthetic wave is 40 pixels high. One central row cycles model, cost band, thinking effort, branch, then project; unavailable values are skipped using the existing fade/scroll style. There is no fixed cost pill. The lower hint alternates Shake to fade and We will send the audio to Codex for transcription in the same position; the long message scrolls inside the clipped row. Show both elapsed and remaining seconds. Native endpoint attachment, account/backend, dependencies, installation, cryptographic wire checks, device stack headroom and physical capture/delivery remain unverified. Whole native deadline: 15 seconds; uncertainty never becomes success. Evidence / limits: PDF generation, rendering and visual review only. No tests, scans, configure/build/size, simulator, flash, hardware or native-input execution.
2026-10-12 CHANGE	**Audio review: bounded window, readiness and honest uncertainty** Affected: Current main pages 24-26; original voice screens 28-30; firmware voice owner and PC native-audio broker Replace the one-packet DATA window with four 480-sample packets. Cumulative authenticated ACK names the highest contiguous sequence; out-of-order DATA is discarded while that prefix is acknowledged. Retry identical ciphertext after 100 ms. Transport runs every 5 ms with at least one scheduler tick. Ten-second unacknowledged PSRAM/grace may survive CCP disconnection only for the same authenticated key and IP. Transport failure after transmitted END remains Unconfirmed. Add a periodic read-only native initialize/thread-read probe: 5-second whole budget, refresh every 5 seconds, freshness 10 seconds. START requires recent readiness matching the genuine followed chat. Readiness does not certify localAudio, model support or transcription. Keep the START chat frozen, then resolve its current hook turn at submission for expectedTurnId. A concurrent native turn conflict fails explicitly; keep the WAV and do not retry automatically. Use the recording hex ID as clientUserMessageId for correlation without promising idempotency. Status exposes recording ID, native turn, WAV artifact and uncertainty. These source changes supersede the earlier single-packet and START-frozen-turn description without erasing that dated history. Evidence / limits: Source implementation and PDF authoring only; tests, build, scans, installation and physical capture/delivery remain unverified.

Revision register / Security

Original dates, screen IDs, decisions and evidence are retained. Source code does not prove device behavior.

DATE / TYPE	DESCRIPTION / ORIGINAL PAGE AND SCREEN IDS
2026-10-10 SECURITY	**Fresh endpoint-bound proof before current data** Affected: Core page 4; FIND / READY / COMMITTED; screens 12-15 and current activity/questions A captured authenticated FIND can request a challenge, but cannot establish presence, change the current notification destination or obtain ACTIVITY, TECH or EVENT. READY is signed over device, pc, nonce, a random challenge, observed device IPv4 address and UDP port. Cody verifies that endpoint and answers COMMITTED. The ready-v2 and committed-v2 authentication domains require the matching fresh one-time response from that endpoint before commit recovery, liveness and current-data delivery. This is a coordinated firmware/plugin update; no old-handshake fallback is provided. Evidence / limits: CCP 0.6.3 source implementation/static review only; no deployment, pairing or adversarial runtime evidence accompanies this PDF.
2026-10-10 SECURITY	**Bound unauthenticated work and retained logs** Affected: Core page 10; device worker, PC discovery/pairing, broker ingress and runtime logs Firmware limits each UDP receive pass by packet count and time, then runs local commands and protocol maintenance even while traffic remains queued. Fixed global and per-source ingress budgets precede expensive authentication. Rotating source addresses cannot bypass the global budget. PC discovery bounds total packets/offers and per-source offers; verifier work also has a time/work ceiling. Overflow is an explicit failure with no association staged, not an incomplete result labelled complete. Broker admission precedes credential work; rejection summaries and the exclusively owned rotating broker log bound network diagnostic growth. Residual: startup-only runtime.log still appends across launches. LAN UDP rejection traffic cannot grow that bootstrap file; bounding repeated-launch output is deferred separately from the implemented broker/network log rotation. Evidence / limits: Source-only limits need measurement against real pairing/reconnection bursts. No suite, flood probe, build or flash is claimed.
2026-10-10 SECURITY	**Remaining protocol risks are explicit** Affected: Pairing verifier/link-key derivation; ACTIVITY / TECH / EVENT; optional HTTP Preserve local pairing admission, expiring codes, constant-time proofs, distinct active/pending journals, strict packet/identity checks, signed choices/settings/debug/reset and loopback-authenticated PC control. Native approval stays on the PC; storage/authentication failures remain failures. Six-character pairing transcripts still permit offline candidate checking; rate limiting does not solve it. CCP payloads remain authenticated and unencrypted. Standard PAKE or high-entropy QR admission followed by authenticated encryption is deferred as a coordinated redesign. An enabled optional device HTTP service needs its own review. Evidence / limits: No confidentiality or production-security certification. Compilation, stack headroom and physical recovery remain pending.